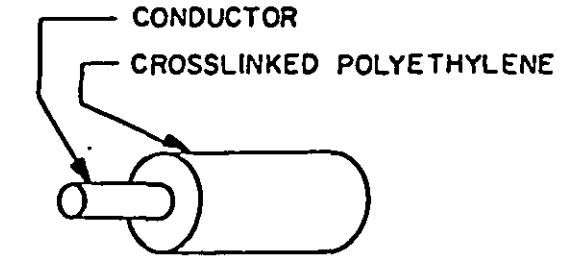
MILITARY SPECIFICATION SHEET

# WIRE, ELECTRICAL, CROSSLINKED, MODIFIED POLYETHYLENE (XLPE) INSULATED, 125°C, 3000 VOLTS

This specification sheet is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the wire described herein shall consist of this document and the latest issue of MIL-W-16878.




SH 11880

FIGURE 1. Construction.

TABLE I. Construction details.

| Wire size | Stranding | Conductor | | Conductor diameter (nominal) (inch) | Finished wire | |
|---|---|---|---|---|---|---|
| | | | | | Diameter (inch) | |
| | | Material | Coating | | Min | Max |
| 26 | 1 X 26 | Copper | Tin | 0.0159 | 0.067 | 0.082 |
| 26 | 7 X 34 | Copper | Tin | .019 | .070 | .085 |
| 26 | 19 X 38 | Copper | Tin | .020 | .070 | .085 |
| 24 | 1 X 24 | Copper | Tin | .0201 | .071 | .086 |
| 24 | 7 X 32 | Copper | Tin | .024 | .075 | .090 |
| 24 | 19 X 36 | Copper | Tin | .025 | .075 | .090 |

(A) denotes changes.

FSC 6145



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TABLE I. Construction details. - Continued

| Wire size | Stranding | Conductor | | Conductor diameter (nominal) (inch) | Finished wire Diameter (inch) | |
|---|---|---|---|---|---|---|
| | | Material | Coating | | Min | Max |
| 22 | 1 X 22 | Copper | Tin | 0.0254 | 0.076 | 0.091 |
| 22 | 7 X 30 | Copper | Tin | .030 | .081 | .098 |
| 22 | 19 X 34 | Copper | Tin | .032 | .081 | .098 |
| 20 | 1 X 20 | Copper | Tin | .0320 | .083 | .098 |
| 20 | 7 X 28 | Copper | Tin | .038 | .089 | .105 |
| 20 | 19 X 32 | Copper | Tin | .040 | .089 | .105 |
| 18 | 1 X 18 | Copper | Tin | .0403 | .091 | .106 |
| 18 | 7 X 26 | Copper | Tin | .048 | .099 | .114 |
| 18 | 19 X 30 | Copper | Tin | .047 | .099 | .114 |
| 16 | 1 X 16 | Copper | Tin | .0508 | .102 | .117 |
| 16 | 19 X 29 | Copper | Tin | .057 | .108 | .123 |
| 16 | 26 X 30 | Copper | Tin | .060 | .111 | .125 |
| 14 | 1 X 14 | Copper | Tin | .0641 | .115 | .130 |
| 14 | 19 X 27 | Copper | Tin | .072 | .123 | .138 |
| 14 | 41 X 30 | Copper | Tin | .076 | .127 | .142 |
| 12 | 19 X 25 | Copper | Tin | .092 | .156 | .171 |
| 12 | 37 X 28 | Copper | Tin | .089 | .153 | .168 |
| 12 | 65 X 30 | Copper | Tin | .093 | .157 | .172 |
| (A) 10 | 37 X 26 | Copper | Tin | .111 | .174 | .189 |
| 8 | 133 X 29 | Copper | Tin | .167 | .239 | .254 |
| 8 | 168 X 30 | Copper | Tin | .169 | .239 | .254 |
| 6 | 133 X 27 | Copper | Tin | .210 | .283 | .298 |
| 4 | 133 X 25 | Copper | Tin | .266 | .346 | .366 |
| 4 | 133 X 25 | Copper | Tin | .268 | .346 | .366 |
| 2 | 133 X 23 | Copper | Tin | .355 | .415 | .435 |
| 2 | 665 X 30 | Copper | Tin | .342 | .415 | .435 |
| 1 | 259 X 25 | Copper | Tin | .378 | .468 | .488 |
| 1 | 817 X 30 | Copper | Tin | .382 | .468 | .488 |
| 0 | 259 X 24 | Copper | Tin | .424 | .518 | .538 |
| 0 | 1045 X 30 | Copper | Tin | .431 | .518 | .538 |
| 00 | 1330 X 30 | Copper | Tin | .475 | .575 | .599 |
| 000 | 1672 X 30 | Copper | Tin | .535 | .657 | .681 |
| 0000 | 2109 X 30 | Copper | Tin | .600 | .709 | .733 |

REQUIREMENTS:

Visual and mechanical inspection: Applicable.
Spark test: 8.0 kilovolts (kV).
Impulse dielectric test: 12.0 kV.
Tank test: 6.0 kV.
Dielectric withstanding voltage: 6.0 kV.
Insulation resistance: $IR = K \log_{10} D/d$



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Where: IR = Minimum insulation resistance in megohms per 1000 feet at 20°C
K = 10,000
D = Maximum average diameter of finished wire
d = Conductor diameter

Cold bend: Condition 4 hours at minus 55 ± 1°C (see table II).

Ⓐ TABLE II. Cold bend mandrel sizes.

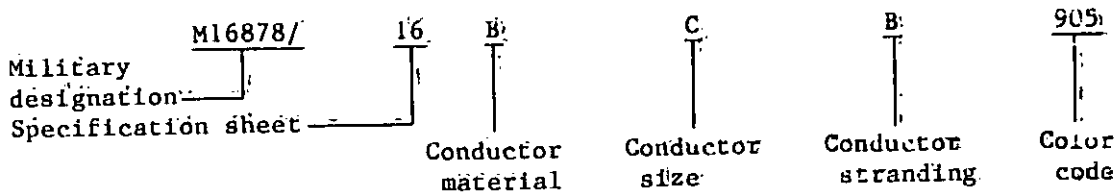
| Wire size | Cold bend mandrel (maximum diameter) (inches) |
|---|---|
| 26 thru 22 | 2 |
| 20 thru 14 | 3 |
| 12, 10 | 4.5 |
| 8 | 6 |
| 6, 4 | 10 |
| 2, 1, 0 | 18 |
| 00, 000, 000 | 20 |

Concentricity: 70 percent (minimum).
Surface resistance: 5 megohms (minimum).
Wrap back: Not applicable.
Heat resistance: Condition at 150°C.
Flammability: Applicable - 30 seconds (maximum) burning time.
3 inches (maximum) flame travel.
Soldering: Applicable.
Heat aging: 25 percent change (maximum) 96 hours at 135°C.
Insulation tensile strength: 1800 pounds force per square inch (minimum).
Insulation elongation: 100 percent (minimum).
Stripe durability: Applicable.
Fungus resistance: Applicable.
Maximum direct current resistance of finished wire:
See table I of MIL-W-16878.



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Military part number (see MIL-W-16878):




Review activities:
Navy - AS, EC

User activities:
Navy - OS, MC

Preparing activity:
Navy - SH
(Project 6145-N305)



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